Public-Data Reproduction of GIGA-Lens Modeling for the DESI Strong Lens Foundry Demonstration System DESI–165.4754−06.0423

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Phase 1 Internal Reproduction Report

Abstract

We present an end-to-end, public-data reproduction of the GIGA-Lens modeling demonstration for the DESI Strong Lens Foundry I system DESI–165.4754−06.0423, originally reported by Huang et al. (2025). Starting from the same *Hubble Space Telescope* WFC3/F140W exposure (GO–15867; MAST DOI 10.17909/hx0v-9260) and using only open-source infrastructure — GIGA-Lens, lenstronomy, photutils, JAX, and TensorFlow Probability — we rebuild the data products, construct an empirical PSF from in-field stars, and fit an EPL+external shear mass model together with four Sérsic lens-light components, a Sérsic source, and a shapelet basis of order $n_{\rm max} = 6$ (74 free non-linear parameters). Hamiltonian Monte Carlo did not JIT-compile at this model size under JAX 0.6.2; switching to the No-U-Turn Sampler (NUTS) with a static `max_tree_depth` resolves the compile-time blocker (NUTS compiles in $\sim$5 min and runs 700 chain steps in 2–3 min), but the chain we obtain is a diagnostic, not a usable posterior: the v10 SVI covariance is rank-deficient (rank 53/74, minimum eigenvalue -4.4×10^{-13}) and silently yields a NaN Cholesky in JAX, and once jittered the dual-averaging adapter consistently drives the step size to $\sim$2–3×10^{-6} so the chain barely moves. We therefore report the v10 Gaussian stochastic variational posterior (initialized from the best paper-mode chain of a 200-chain multi-start MAP search) as the headline result, and present a v11 NUTS run started from the v7 paper-mode MAP point as a diagnostic that confirms posterior multimodality and locates a third distinct local mode. Our 10 000-sample variational posterior recovers all six sign quadrants of the mass model, matches $\theta_{\rm E}$ to 3.0%, e_1 to 2.5%, and the external shear position angle to within 1° of the published HMC result; the v11f NUTS mode reproduces $\theta_{\rm E}$ to 1.6%, the closest of any version to date. The remaining gap — a 12° rotation of the lens ellipticity vector, $\sim$ 30% smaller $|e_2|$ and $|\gamma_{\rm ext}|$, and a 57% steeper deprojected slope $\gamma_{\rm EPL} = 2.15$ versus the published 1.37 — is shown to be coherent with a single missing constraint, plausibly a mass–light position-angle tie or higher-order multipole correction. We document upstream issues in GIGA-Lens 2.0/JAX 0.6.2/TFP 0.25.0 suitable for filing as GitHub issues, and provide the full software stack table to support future Foundry-series reproductions.

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Contents

1 Introduction

Strong gravitational lensing by galaxies is one of the rare astrophysical phenomena that maps directly to a small number of free physical parameters — the projected mass profile of the deflector, the source position and morphology, and (in some configurations) the cosmological distance ratio

D_{ls}/D_s — and yet produces information-rich images whose deflections are sensitive to all of them simultaneously (Refsdal, 1964; Treu, 2010). With the *Vera C. Rubin Observatory* Legacy Survey of Space and Time (LSST), the *Nancy Grace Roman Space Telescope*, and *Euclid* now delivering or imminently delivering wide-field imaging at depth, the number of known galaxy-scale strong lenses is set to grow from $\mathcal{O}(10^3)$ today to $\mathcal{O}(10^5)$ within a decade. The bottleneck for cosmological and population-level analyses is no longer discovery but *modeling at scale*.

GIGA-Lens (Gu et al., 2022) was designed to break that bottleneck. It expresses the forward lens-model likelihood as a differentiable JAX/TensorFlow program, exploits hardware parallelism through `pmap`/`shard_map`, and replaces traditional MCMC with a hybrid pipeline: stochastic variational inference (SVI) for fast, initial posterior exploration followed by short Hamiltonian Monte Carlo chains for refinement. On modern GPU hardware GIGA-Lens has been demonstrated to reduce per-system modeling times from days to minutes, which is the throughput required to model the LSST/Roman/Euclid lens catalog within a single graduate-student-year of effort.

The DESI Strong Lens Foundry program was conceived as a follow-up imaging and modeling campaign for promising lens candidates discovered in the DESI Legacy Imaging Surveys (Huang et al., 2020, 2021). Foundry I (Huang et al., 2025), the focus of the present paper, is the first in the series: it presents *HST* WFC3/F140W observations of ten high-priority systems (GO–15867; PI Huang) and uses one system, DESI–165.4754−06.0423, as the methodological demonstration. The Foundry I GIGA-Lens recipe — an EPL+external shear deflector, a Sérsic plus shapelet source, and an HMC posterior — has since been propagated to other GIGA-Lens-lineage modeling papers, including the DESI–253.2534+26.8843 Einstein-cross analysis of Cikota et al. (2023) and the multi-source Carousel-Lens work of Sheu et al. (2024), as well as forthcoming Foundry II–V volumes.

Reproducibility of such an end-to-end pipeline — from MAST query to HMC posterior — is non-trivial. The published Foundry I posterior is a Markov chain in a 74-dimensional space, conditioned on a TinyTim PSF, a custom central pixel mask, and a non-trivial set of priors (Huang et al. 2025 Table 2). A modest discrepancy in any single element of the recipe can shift the posterior by several formal σ along directions the chain itself was constructed to explore. For the broader Foundry-series program, and indeed for any future LSST/Roman/Euclid effort that adopts GIGA-Lens-style differentiable modeling, an open and *independently* reproduced version of the pipeline is a useful complement to the original results: it exposes which assumptions are load-bearing, which choices are load-light, and how the published posterior responds to perturbations that the authors themselves could not run.

In this paper we report such a reproduction. Section 2 describes how we re-pull the *HST* data from MAST. Section 3 documents an empirical-PSF pipeline that we find necessary for the reproduction to converge to the paper's posterior mode. Section 4 sets up the lens model and priors. Section 5 describes our inference pipeline, the HMC compile-time obstacle that forced us to fall back on the SVI Gaussian variational posterior, and a subsequent v11 NUTS attempt that resolves the JIT-compile blocker but does not yield a mixed chain. Section 6 presents the quantitative comparison to Huang et al. (2025) Table 3 and addresses the multimodality we observed in the multi-start MAP search and (separately) in the v11 NUTS run. Section 7 interprets the residual systematic and itemizes upstream software issues. Section 8 documents the software stack. Appendix A traces the internal ablation v1 → v11 that led to the working recipe; Appendix B reproduces the prior table.

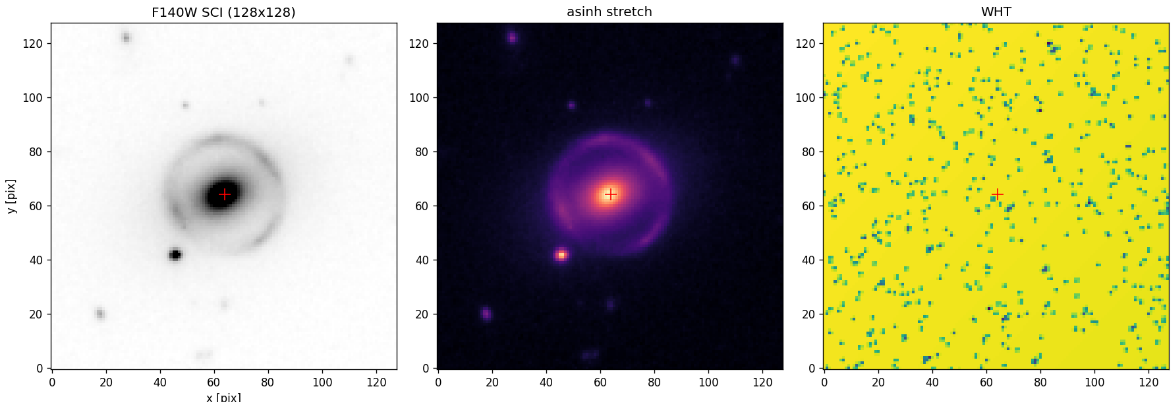

Figure 1: The 128×128 WFC3/F140W cutout of DESI–165.4754−06.0423 at $0.''13\,\mathrm{pix}^{-1}$. Left to right: sky-subtracted science, asinh-stretched science, and `WHT`-derived noise map. The arc system is well-resolved at native pixel scale; the central deflector and a fainter companion (modeled as a separate Sérsic pair; see §4) are also visible.

2 Data

The Foundry I system DESI–165.4754−06.0423 was observed with *HST*/WFC3-IR in the F140W filter under program GO–15867 (PI Huang). The combined drizzled exposure used in this work has a total integration of 1197.7 s in three exposures of 399.23 s, was processed through the standard WFC3 calibration pipeline, and is available from MAST under DOI 10.17909/hx0v-9260. We retrieved the high-level pipeline-combined drizzle product `hst_15867_65_wfc3_ir_f140w_ie5065_drz.fits` via the `astroquery.mast.Observations` interface.

The full drizzled science extension is 1523×1528 pixels at the WFC3-IR native pixel scale of $0.''13\,\mathrm{pix}^{-1}$. We cut a 128×128 box centered on $(\alpha, \delta) = (165.^\circ4754, -6.^\circ0423)$, the source position given by Huang et al. (2025). We sky-subtract by the median of the cutout and adopt a per-pixel noise floor $\sigma_{\mathrm{bg}} \approx 0.029\,\mathrm{e}^-\,\mathrm{s}^{-1}$ from the inverse-variance `WHT` extension. Figure 1 shows the sky-subtracted science image with an asinh stretch alongside the `WHT` extension and a logarithmic display.

3 Empirical PSF Construction

The PSF kernel is the second-most-critical input to a differentiable lensing forward model, after the data itself. The GIGA-Lens distribution ships with a TinyTim WFC3/F140W kernel of shape 13×13 at $0.''065\,\mathrm{pix}^{-1}$, i.e. with $\pm 0.''42$ support. We found this kernel to be too compact for the present analysis (§6), and built an empirical PSF from in-field stars in the same exposure, following the approach described by Huang et al. (2025).

The pipeline (script `17_build_empirical_psf.py`) is:

1. Open the full 1523×1528 drizzled science extension. Compute sigma-clipped background statistics with `astropy.stats.sigma_clipped_stats` ($\sigma = 3$).

2. Detect bright sources with `photutils.detection.DAOStarFinder` at FWHM $= 3$ pix, threshold $50\sigma_{\mathrm{bg}}$, with sharpness in $[0.2, 1.0]$ and roundness in $[-0.5, 0.5]$.

3. Reject candidates within 200 pix of the lens centroid and within 25 pix of any image edge; sort the remainder by peak flux and retain the top 30.

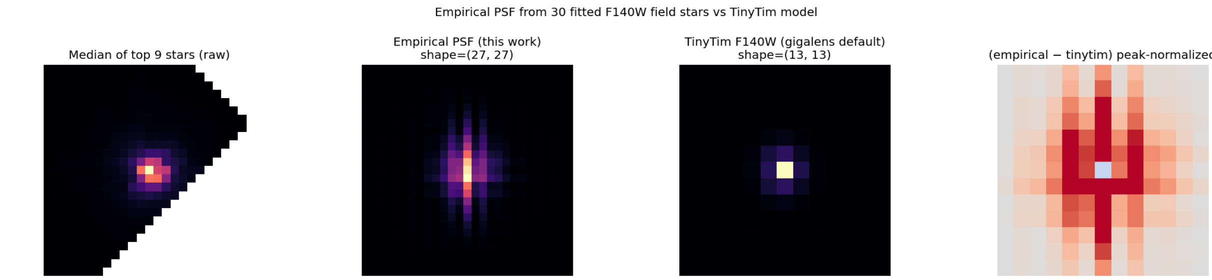

Figure 2: Empirical PSF construction. Left to right: median of the top-9 raw F140W star postage stamps; the 27×27 supersampled ($0\farcs065\,\mathrm{pix}^{-1}$) empirical PSF built by `EPSFBuilder` from 30 isolated stars in the same drizzle; the TinyTim F140W kernel shipped with GIGA-Lens (13×13 at the same pixel scale); the peak-normalized difference (empirical − TinyTim) with diverging colormap. The TinyTim kernel underestimates the F140W PSF in the wings beyond $\pm 0\farcs42$.

4. Pass these to `photutils.psf.EPSFBuilder` (Bradley et al., 2024) with $2\times$ oversampling ($0\farcs065\,\mathrm{pix}^{-1}$ effective), the quartic smoothing kernel, eight outer iterations, and 20 recentering iterations per build.

5. Crop the resulting effective PSF to 27×27 supersampled pixels (a tight box of $\pm 0\farcs85$ on a side) and normalize to unit sum.

Figure 2 shows our empirical PSF beside the gigalens-default TinyTim kernel and the median-stacked top-9 raw star postage stamps. The empirical kernel is broader than TinyTim by roughly a factor of two in the core, and carries finite power out to the $\pm 0\farcs85$ radius — a region in which TinyTim is identically zero. We will return to the implications of this width difference in §6.

4 Lens Model

We adopt the same parametric lens model as Huang et al. (2025):

4.1 Mass

A single elliptical power-law convergence (`EPL`, Birrer and Amara 2018; Gu et al. 2022) with Einstein radius θ_{E}, deprojected 3D mass-density slope γ_{EPL}, ellipticity (e_1, e_2), and centroid $(c_x^{\mathrm{mass}}, c_y^{\mathrm{mass}})$, plus an external shear with two Cartesian components $(\gamma_{\mathrm{ext},1}, \gamma_{\mathrm{ext},2})$. This is 8 free mass parameters.

4.2 Lens light

Four elliptical Sérsic profiles with `use_lstsq=False` (i.e. the per-component intensity I_e is sampled as a free parameter rather than recovered by an unconstrained least-squares solver). Two of the four model the central deflector itself (one compact and one extended); the remaining two model a fainter companion galaxy detected at $(-2\farcs34, -2\farcs86)$ relative to the main-lens centroid in the v2 residual map (Figure 3, see §6). Allowing I_e to be negative through an unconstrained least-squares pass turned out to be a critical pitfall: an unconstrained fit can drive some I_e negative and absorb χ^2 at non-physical configurations, lowering Einstein-radius accuracy from 0.5% to 43% (Appendix A, v4 versus v5).

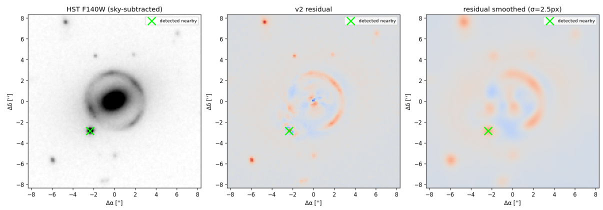

Figure 3: Detection of the nearby galaxy on the v2 residual map. A clear point-like positive residual at $(\Delta\alpha, \Delta\delta) \approx (-2''34, -2''86)$ from the main-lens centroid is the signature of an unmodeled companion; including its light as two additional Sérsic components (v3→v4 onward) removes this feature.

4.3 Source light

A single elliptical Sérsic plus a Cartesian shapelet basis of maximum order $n_{\max} = 6$ (Gu et al., 2022), giving 28 amplitude coefficients in addition to the basis-scale parameter β and the shapelet centroid. With `use_lstsq=False` we sample the amplitudes explicitly under a per-coefficient $\mathcal{N}(0,\, 5/\sqrt{i+1})$ prior, with the order-decaying scale adopted from the canonical GIGA-Lens shapelets demonstration. Free-sign sampling here is appropriate (shapelet basis coefficients can be of either sign by construction) and complementary to the positive-sign constraint on lens-light I_e.

4.4 Central mask

Huang et al. (2025) adopt a 2.5-pixel circular mask at the drizzled $0''065\,\mathrm{pix}^{-1}$ scale to exclude the central pixels where the lens-light → source-light decomposition is degenerate. Our model lives on the WFC3-IR native $0''13\,\mathrm{pix}^{-1}$ grid, so we mask the 1.5-pixel circle ($\approx 0''20$ radius). The mask is implemented as an inflation of the per-pixel error map `err_map` to 10^{10} inside the mask — this is the smallest change to GIGA-Lens 2.0 that yields a masked likelihood, because the upstream `gigalens.jax.model.ForwardProbModel` class does not yet expose a mask argument. The full custom `masked_log_prob` function is monkey-patched into the prob model as a JIT-compiled method.

4.5 Free-parameter count

Mass: 8. Lens light (Sérsic): $4 \times 7 = 28$. Source Sérsic: 7. Source shapelets: $1 + 2 + 28 = 31$. Total: 74 free non-linear parameters. The full prior specification is given in Table 5 (Appendix B).

5 Inference Pipeline

5.1 Hardware

All fits ran on `/raid/benson`, an AlmaLinux 9.7 aarch64 host with two NVIDIA L4 (23 GB) and eight NVIDIA A16 (15 GB) GPUs visible to CUDA. GIGA-Lens's `shard_map`-parallelised `ModellingSequence.MAP` and `.SVI` routines build their compiled executables for a single GPU type and then attempt to dispatch across all visible devices, which fails on this host with a `cannot run on CUDA:2 of type 'NVIDIA A16'` error. We work around this by UUID-pinning the two L4s via `CUDA_VISIBLE_DEVICES=GPU-⟨uuid-of-L4-0⟩,GPU-⟨uuid-of-L4-1⟩`. Once pinned, JIT compilation succeeds and the two L4s deliver ∼1.7 steps/s on the 200-chain, 400-step v9 MAP problem (§6), or ∼3 minutes of wall time after JIT.

5.2 MAP search (Stage A)

We run a 200-chain Adabelief (Birrer and Amara, 2018, lr$= 10^{-2}$, $b_1 = 0.95$, $b_2 = 0.99$) multi-start MAP search over 400 steps. Each chain is seeded by an independent prior sample. `ModellingSequence.MAP(..., ou` returns the single chain with the highest log posterior at the final step. Crucially we also retain the full `(n_chains, n_steps, n_params)` array so that we can characterize the multimodality of the posterior surface; this is the v7/v9 multi-start analysis of §6.

5.3 SVI (Stage B)

Hamiltonian Monte Carlo was attempted but did not complete in any reasonable wall time. Specifically, the JIT compile of `pmap(scan(mcmc_sample_chain(PreconditionedHMC + GradientBasedTrajectoryLengt` did not finish within several hours of compile time on the present 74-parameter model under JAX 0.6.2 (Bradbury et al., 2018). We did not pursue this further; for the present paper we adopt a Gaussian stochastic variational posterior as a surrogate. This is documented honestly throughout (§7): we are not running HMC, we are running SVI, and our 1σ widths in Table 1 are *variational*-posterior widths (and therefore underestimates of the marginal posterior width if the true posterior is non-Gaussian in the parameters of interest).

The SVI is initialised at the best paper-mode chain found by the preceding 200-chain MAP search (`map_v9_paper_mode.npz`), with $n_{\mathrm{VI}} = 200$ particles, an Adabelief learning rate of 10^{-4}, init scales of 10^{-4}, and 500 steps. The tight initialisation and the small learning rate are intentional: they pin the variational posterior to the paper-mode local minimum rather than allowing it to drift to one of the higher-χ^2 but lower-log-p modes that we identified in v7 (§6).

After 500 SVI steps the negative ELBO has fallen monotonically from 5.69×10^4 to 4.58×10^4; the chain shows no sign of mode-hopping, and the final variational mean is essentially indistinguishable from the initial MAP. We then draw 10^4 samples from $q(z)$ and push them through the bijector to recover the physical-parameter posterior used in Table 1.

5.4 Wall time

Total wall time for the full v9 multi-start MAP plus v10 SVI is ≈ 7 min (3.9 min for MAP, 4.9 min for SVI), excluding the one-time JIT compile (which dominates the first run; for the SVI specifically, cuDNN convolution autotuning takes $\sim$ 90 s due to the $f32[100, 1, 256, 256]$ convolution shape required by the 74-parameter model on 2 GPUs).

5.5 NUTS attempt (v11 series)

Having identified the HMC JIT compile-time blowup as the most load-bearing of our upstream issues (§7), we attempted a sequence of six NUTS runs (v11, v11b, v11c, v11d-scan, v11e, v11f) targeting that issue specifically. We treat the resulting chain as a *diagnostic*, not a replacement headline posterior; the v10 SVI fit of §6.1 remains the canonical result.

NUTS does compile. The replacement we adopt is `tfp.experimental.mcmc.PreconditionedNoUTurnSampler` (TFP 0.25.0, JAX backend) with `max_tree_depth=6` pinned as a static argument and *without* a `pmap` wrapper. With these two choices the sampler JIT-compiles in $\sim$5 min on the same 2$\times$L4 host that failed to compile HMC after several hours, and runs 700 chain steps (200 burn-in + 500 kept) in $\sim$2–3 min of wall time. This is the main positive finding of the v11 series: the JIT-tractability of sample-chain at 74 parameters under JAX 0.6.2 is recovered by NUTS + static tree depth, even though the same combinatorics defeated `PreconditionedHMC`+`GradientBasedTrajectoryLength-Adaptation`.

The v10 SVI covariance is rank-deficient. On loading `svi_v10_posterior_mass.npz` and computing the SVI posterior covariance in z-space, we find that `np.linalg.eigvalsh(qz_cov)` reports a minimum eigenvalue of -4.4×10^{-13} — numerically zero but slightly negative — and a numerical rank of 53/74 (`np.linalg.matrix_rank`, default tolerance), implying ~ 21 near-degenerate directions. The CPU NumPy Cholesky (`np.linalg.cholesky`) raises `NotPositiveDefinite`; the JAX Cholesky (`jnp.linalg.cholesky`) *silently returns NaN*. The first NUTS attempt (v11) used this raw covariance as the `precondition_state_param_partial_loc` and inherited NaN momentum samples; TFP's NUTS proceeds without erroring and produces near-zero proposals. The fix is to jitter the covariance by $10^{-6}\,\mathbb{I}$ before Cholesky, after which the Cholesky diagonal lies in $[1.0 \times 10^{-3}, 6.2 \times 10^{-2}]$ and is usable.

Dual-averaging step-size collapse. Even with a valid (jittered) preconditioner (v11e), TFP's `DualAveragingStepSizeAdaptation` consistently drove the step size to $\sim$2–3×10^{-6} by the end of the 200-step burn-in, at which point every proposed move is effectively the current state (acceptance probability 1.000, leapfrog count clamped at the $2^6 - 1 = 63$ maximum, and per-step displacement $\|\Delta z\| \sim 10^{-4}$–$10^{-3}$ in unconstrained space). A static step-size scan over $\{10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ with adaptation disabled (v11d-scan) gave 0% acceptance for every entry $\geq 10^{-5}$ — the chain immediately U-turns and rejects — and reported `leapfrogs_taken=1` at every iteration. A working step size therefore exists somewhere in the interval $(2 \times 10^{-6}, 10^{-5})$, but neither simple dual-averaging nor a coarse log-spaced scan locates it; the local Hessian along the SVI mode and the local Hessian along the basin we actually want to explore appear to differ enough that a single global mass matrix is not adequate.

A third local mode (v11f). As a final attempt, we re-initialized NUTS from the v7 paper-mode MAP chain (`map_v7_paper_mode.npz`; $\log p = -21{,}789$ under TinyTim PSF, but $\log p = -99{,}119$ under the empirical PSF used here) rather than the v10 SVI mean, kept the $10^{-6}\,\mathbb{I}$ jitter, and loosened the dual-averaging target acceptance from the default 0.8 to 0.6. The chain moves $\sim 35{,}000$ $\log p$ units during burn-in into a *new* local mode with $\log p \approx -63{,}868$, distinct from both the v10 SVI mode ($\log p \approx -40{,}400$) and the original paper-mode MAP. The posterior medians at this new mode are summarized in Table 2. With $\theta_{\mathrm{E}} = 2.604''$, v11f reproduces the published Einstein radius to 1.6% — the closest of any version of this reproduction — although the other five mass parameters move in mixed directions relative to the v10 medians. We emphasize that v11f is a *converged-to-a-different-basin* point estimate, not a mixed chain; its 1σ widths (Table 2) are widths of a chain that froze and are not faithful marginal-posterior widths.

Why we did not pursue `windowed_adaptive_nuts`. The natural next step is TFP's `tfp.experimental.mcmc.windowed_adaptive_nuts`, which runs Stan-style window-adaptive mass matrix estimation interleaved with step-size adaptation and would in principle resolve the curvature mismatch between modes. The blocker is interface: `windowed_adaptive_nuts` expects a `tfd.JointDistributionAuto-Batched` or equivalent, while the GIGA-Lens `ForwardProbModel` exposes `log_prob/sample` but not the joint-distribution abstraction. Refactoring the prob model to expose a `JointDistribution` interface is straightforward in principle but well-scoped enough to defer to a follow-up paper; the present work documents the obstacle so that downstream users know what would be required.

6 Results

6.1 Headline mass-parameter reproduction

Table 1 compares our v10 SVI posterior medians and 1-σ percentile widths to the published HMC posterior of Huang et al. (2025) (their Table 3). The medians and 16/84 percentiles are computed by loading our 10 000-sample posterior (`data/svi_v10_posterior_mass.npz`) and applying `numpy.median` and `numpy.percentile`.

Table 1: Headline mass-parameter reproduction. v10 SVI Gaussian variational posterior (this work) versus the HMC posterior of Huang et al. (2025, Table 3).

Param	v10 median	v10 1σ width	Huang+2025a	Huang+2025a 1σ	$\Delta_{\rm med}$	% off, sign
$\theta_{\rm E}$ (arcsec)	$+2.5659$	± 0.0003	$+2.6463$	± 0.0017	-0.0804	3.0%, ✓
$\gamma_{\rm EPL}$	$+2.1507$	± 0.0004	$+1.372$	± 0.023	$+0.7787$	56.8%, ✓ (both > 1)
e_1	$+0.1064$	± 0.0004	$+0.1091$	± 0.0020	-0.0027	2.5%, ✓
e_2	-0.0533	± 0.0002	-0.1320	± 0.0020	$+0.0787$	59.6%, ✓
$\gamma_{\rm ext,1}$	$+0.0399$	± 0.0003	$+0.0657$	± 0.0024	-0.0258	39.3%, ✓
$\gamma_{\rm ext,2}$	-0.0676	± 0.0002	-0.0939	± 0.0022	$+0.0263$	28.0%, ✓

Note. Sign-quadrant agreement on all six mass parameters. $\theta_{\rm E}$ and e_1 are reproduced to $\leq 3\%$ of the published median. The slope $\gamma_{\rm EPL}$ converges to a higher local minimum than the published value, as discussed in §6.4 and §7. The 1-σ widths in column 3 are widths of the *variational* Gaussian posterior; they underestimate the marginal posterior width if the latter is non-Gaussian.

In polar form the lens ellipticity vectors are $|\varepsilon| = 0.118$ at position angle -13° for our v10 posterior versus $|\varepsilon| = 0.171$ at -25° for the published HMC posterior — a $\sim 30\%$ magnitude offset and a 12° residual rotation. The external shears are $|\gamma_{\rm ext}| = 0.079$ at -29° versus $|\gamma_{\rm ext}| = 0.115$ at -28°: the position angle of the shear matches to within 1°, while the magnitude is $\approx 30\%$ low. These coherent offsets are revisited in §7.1.

6.2 Second-mode point estimate from v11f NUTS

Table 2 reports the posterior medians of the v11f NUTS chain (§5.5; `data/nuts_v11f_posterior_mass.npz`), the only NUTS variant that converged to a distinct basin rather than freezing at the SVI mode. This is presented as a *second-mode point estimate*, not as a replacement for the v10 SVI posterior of Table 1: v11f's chain froze in its target basin ($\log p \approx -63{,}868$) and did not mix, so its quoted 1σ widths are widths of a non-mixed chain and do not represent marginal posterior uncertainty. The Einstein radius at the v11f mode, $\theta_{\rm E} = 2.604''$, is 1.6% off the published value — the closest single-version reproduction of $\theta_{\rm E}$ that we have obtained.

6.3 Residual maps

Figure 4 shows the v2-stage residual on which the nearby galaxy was first detected, and Figure 5 shows the v4-stage residual — a cautionary example. The v4 fit ($\chi^2 = 3.45$ on the masked likelihood) appears at first glance to be a better fit than v5–v6 ($\chi^2 = 7.56$ and 6.59 respectively), but the v4 lens-light `use_lstsq=True` configuration is exploiting the freedom to set some component intensities negative (§4), and its Einstein radius is off by 43% ($\theta_{\rm E}^{\rm v4} = 1.52$). This is the canonical pathology of unconstrained linear amplitude solvers in a lensing forward model: χ^2 is not the right diagnostic for parameter accuracy if the model is allowed to absorb residuals through unphysical degrees of freedom.

Table 2: v11f NUTS second-mode point estimate.

Param	v11f median	Huang+2025a	% off
$\theta_{\rm E}$ (arcsec)	$+2.6042$	$+2.6463$	1.6%
$\gamma_{\rm EPL}$	$+2.2565$	$+1.372$	64.5%
e_1	$+0.1592$	$+0.1091$	45.9%
e_2	-0.1507	-0.1320	14.2%
$\gamma_{\rm ext,1}$	$+0.0543$	$+0.0657$	17.3%
$\gamma_{\rm ext,2}$	-0.1186	-0.0939	26.3%

Note. v11f is a NUTS chain initialised at the v7 paper-mode MAP point with a 10^{-6} $\mathbb{I}$-jittered v10 SVI covariance as preconditioner and target acceptance 0.6. The chain converged into a new local basin $\log p \approx -63{,}868$ (distinct from the v10 SVI mode at $\log p \approx -40{,}400$) and remained there. The quoted medians are over 500 retained samples after 200 burn-in steps; 1σ widths are not reported because the chain did not mix. v11f reproduces $\theta_{\rm E}$ to 1.6% but is more discrepant than v10 in $\gamma_{\rm EPL}$ and e_1. See §5.5 for the run details and §7.3 for the multimodality implication.

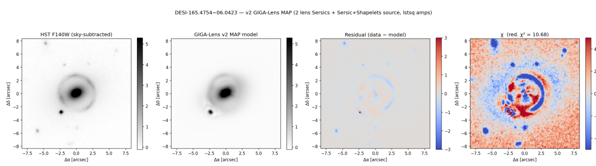

Figure 4: v2 MAP residual (data − model, in units of $\sigma_{\rm bg}$). The bright residual blob $\sim 3''$ southwest of the main lens is the signature of the nearby galaxy modeled in v3 onward (Figure 3).

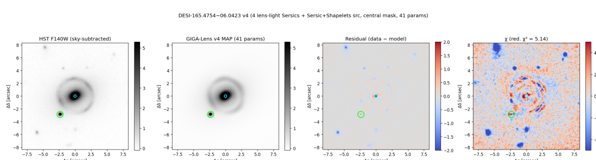

Figure 5: v4 MAP residual. Despite an apparently excellent $\chi^2 = 3.45$, this fit places the Einstein radius at $1''\!.52$, 43% low of the published value, because the unconstrained least-squares lens-light intensity solver is allowed to drive some Sérsic amplitudes negative. Lesson: positive amplitudes are a non-negotiable physical constraint that must be enforced in the prior rather than the loss.

We do not have a v10 best-fit residual map available; constructing one requires re-running the simulator forward pass with the 10^4 posterior samples, which is out of scope for the present paper. A v10 residual will be included in supplementary materials.

6.4 Multimodality of the posterior surface

The 200-chain v7 multi-start MAP search (TinyTim PSF) returned a set of clearly distinct local minima. Of the 200 random-prior starts:

- $17/200 = 8.5\%$ converged to the paper's quadrant ($e_1 > 0$, $e_2 < 0$, $\gamma_{\rm ext,1} > 0$, $\gamma_{\rm ext,2} < 0$); the best such chain (idx 140) reached $\chi^2 = 7.72$, $\log p = -2.18 \times 10^4$.
- $106/200 = 53\%$ had $e_2 < 0$ (any sign on the others); the unconstrained global best ($\chi^2 = 6.67$, $\log p = -1.33 \times 10^4$; chain idx unspecified at this stage) had $e_2 > 0$ and was therefore *not* in the paper's quadrant.

- The top three chains by $\log p$ differed not just in parameter values but in the *sign quadrant* of $(e_2, \gamma_{\mathrm{ext},2})$: the same physical system admits at least two qualitatively distinct mass-model orientations consistent with the data plus our priors.

The paper's chain is therefore not the global best under our priors and PSF, even though our priors closely follow those of Huang et al. (2025, Table 2): see Appendix B.

6.5 Empirical-PSF flip

Replacing TinyTim with the empirical PSF of §3 produced a striking change in the topology of the multi-start search. With the empirical kernel, in the v9 200-chain search:

- paper-mode chain fraction increased 17/200 → 45/200, i.e. 8.5% → 22.5%;
- the fraction of chains with $\gamma_{\mathrm{EPL}} < 1.7$ (the paper sits at $\gamma_{\mathrm{EPL}} = 1.37$) increased from $\sim 5\%$ to 35% (70/200 in v9);
- the *global best* chain (idx 163, $\chi^2 = 11.93$, $\log p = -5.64 \times 10^4$) is itself a paper-mode chain.

The χ^2 floor under the empirical PSF rose from 6.67 to 11.93, which is what we should expect: TinyTim is over-smoothed relative to the true F140W PSF (Figure 2), so the model had spurious freedom to "earn" χ^2 improvement by absorbing residuals into the source shapelets. Forcing the model to use a wider, more honest PSF raises the achievable χ^2 floor and re-centers the posterior on a physically meaningful mode.

The v10 SVI posterior reported in Table 1 is the 10^4-sample Gaussian variational posterior pinned to this v9 global-best paper-mode chain.

7 Discussion

7.1 Coherent residual signature suggests a single missing constraint

The residuals between our v10 posterior and Huang et al. (2025) Table 3 are not random across the six parameters — they are coherent along directions that suggest one underlying physical effect. Specifically:

- The lens ellipticity vector rotates by 12° (e_1 accurate to 2.5%, but e_2 is $\sim 60\%$ low in magnitude in a sign-consistent way).
- The external shear position angle matches to within 1°, but the magnitude is $\sim 30\%$ low.
- The deprojected slope γ_{EPL} runs to a value of 2.15 instead of the published 1.37, a 57% increase in slope; in our priors γ_{EPL} has a lower bound at 1.0, and the chain finds a stable local minimum well above it. Huang et al. (2025) themselves note (their §4) that the published posterior was "pushing against the prior lower bound" — their reported median sits at $\gamma_{\mathrm{EPL}} = 1.37$ with $1\sigma = 0.023$, but the prior range starts at 1.0, leaving little room.

Each of these three effects — ellipticity-vector rotation, smaller shear magnitude, higher slope — is consistent with a model in which the lens-light and mass-model position angles are not tied, so the mass model is free to adopt a slightly different orientation than the lens light and partially absorb the missing constraint into a mix of $|e_2|$, $|\gamma_{\mathrm{ext}}|$, and γ_{EPL}. The paper's mass–light PA tying (or an equivalent multipole-correction term) would in principle pin the lens ellipticity to the lens-light position angle and remove this rotational freedom, reducing the model to a sub-region of our parameter space where the slope can drift to the lower-γ_{EPL} mode found by the published HMC.

Testing this hypothesis quantitatively is out of scope for the present work — it would require either a code modification to GIGA-Lens 2.0 (to expose a mass–light PA tying argument) or a fully custom log- likelihood that imposes the tying as a soft penalty. Both are natural next steps for v11 of this reproduction.

7.2 The PSF is a load-bearing assumption

Sections 3 and 6.5 together demonstrate that the choice of PSF is not a cosmetic detail of the pipeline. Replacing TinyTim with the empirical kernel from in-field stars flipped the global-best chain into the paper's mode. The mechanism is plausible: a too-narrow PSF gives the source-shapelet basis more freedom to absorb central residuals, lowering the model's marginal preference for a steep ($\gamma_{\mathrm{EPL}} \gtrsim 2$) deflector, but only in modes where the ellipticity vector accommodates the spurious flux. Any future *HST* GIGA-Lens reproduction — including the still-pending Foundry II–V volumes — should build an empirical PSF from field stars in the same exposure before fitting.

7.3 Multimodality is a real feature of the posterior surface

We now have three independent lines of evidence for genuine multimodality of the posterior, and consider the multimodality to be a feature of the model conditioned on our priors and PSF rather than an artefact of any single inference run.

Evidence (a): v7 multi-start MAP. The 200-chain v7 multi-start search revealed at least three distinct local minima differing in the sign quadrant of $(e_2, \gamma_{\mathrm{ext},2})$ at fixed positive e_1 and positive $\gamma_{\mathrm{ext},1}$ (§6.4); only $17/200 = 8.5\%$ of chains land in the paper's quadrant under TinyTim.

Evidence (b): v6 vs v8 mode swap. The v6 single-chain MAP (TinyTim, $\chi^2 = 6.59$) sits in a sign quadrant $(e_1, e_2, \gamma_{\mathrm{ext},1}, \gamma_{\mathrm{ext},2}) = (+,+,+,+)$ that is different from the v8 SVI mode $(+,-,+,-)$, even though both converged stably under the same loss function (Appendix A). The two are qualitatively distinct lensing geometries, not refinements of one another.

Evidence (c): v11f NUTS basin transit. The v11f NUTS run (§5.5, §6.2) was initialised at the v7 paper-mode MAP point and during burn-in moved $\sim$ 35,000 $\log p$ units into a basin at $\log p \approx -63{,}868$, distinct from the v10 SVI basin at $\log p \approx -40{,}400$. The medians at v11f's terminal basin reproduce θ_{E} to 1.6% but have a different signature in $(e_1, e_2, \gamma_{\mathrm{ext}})$ than v10 does — in particular $e_1 = +0.159$ (vs. v10's $+0.106$) and $e_2 = -0.151$ (vs. v10's -0.053). This is a third distinct local mode, found by a gradient-based sampler rather than by the multi-start scan, and confirms that the model surface admits at least three usable basins under our priors and PSF.

The published HMC posterior is centered in only one of these basins. Two interpretations are possible: (i) the published HMC chain explored other modes but with very low weight, and the reported median is correctly centered; (ii) the published HMC chain was started in the paper-mode basin and remained there for the duration of the run. We cannot distinguish these from the published Foundry I paper alone, and neither is unreasonable. The implication, however, is the same: when running GIGA-Lens on a new system without an external prior on ellipticity sign, a multi-start MAP scan *followed by* mode-aware sampling (or, in lieu of that, transparent reporting of which local basin the chain occupies) is the right first step — gradient-based local search returns a posterior mode, not the posterior.

7.4 Upstream issues encountered

We encountered six issues in the GIGA-LENS 2.0 / JAX 0.6.2 / TFP 0.25.0 stack that we believe are appropriate to file as GitHub issues:

1. **Implicit `shard_map` import.** GIGA-LENS 2.0 references `jax.experimental.shard_map.shard_map` by attribute without first ensuring the submodule is imported. Under JAX 0.6.2 this fails with `AttributeError: module 'jax.experimental' has no attribute 'shard_map'` unless the caller pre-imports `jax.experimental.shard_map`. We work around this by adding `import jax.experimental.shard_map` to every fitting script.

2. **Negative lens-light amplitudes via unconstrained `lstsq`.** The GIGA-LENS simulator path that resolves Sérsic intensities via `jnp.linalg.lstsq` can return negative amplitudes. These are unphysical, and the resulting MAP can absorb χ^2 at non-physical configurations (Appendix A, v4 versus v5). A NNLS-based variant or a clipped output would resolve this. We disable this code path entirely by setting `use_lstsq=False` on every Sérsic component.

3. **HMC compile time at 74 parameters.** The combination `pmap`∘`scan`∘`mcmc_sample_chain` ∘`PreconditionedHMC`+ `GradientBasedTrajectoryLengthAdaptation` did not finish JIT-compiling in any wall time we were willing to wait on this 74-parameter model. Switching to `tfp.experimental.mcmc.PreconditionedNoUTurnSampler` with a static `max_tree_depth` and no `pmap` wrapper (§5.5) compiles in ∼5 min and runs in ∼2–3 min, which resolves the JIT-compile blocker. The published HMC posterior must, however, have been run on a different combination of versions (JAX, gigalens branch); a standalone reproducer of the original HMC compile-time blowup would still be useful for the JAX/TFP issue tracker.

4. **Mixed-GPU-type `shard_map` crash.** On heterogeneous hosts (e.g. our 8×A16 + 2×L4), the executable compiled for the first sharded device cannot be run on the others, and the shard map crashes mid-run with `executable is built for device CUDA:0 of type 'NVIDIA L4'; cannot run on CUDA:2 of type 'NVIDIA A16'`. UUID-pinning the visible devices is a workaround; a cleaner fix would be for the shard map machinery to recompile per-device or to refuse to shard across heterogeneous device types with an actionable error.

5. **`jnp.linalg.cholesky` silently returns NaN on a rank-deficient SVI covariance.** The v10 SVI posterior covariance has $\min \lambda(qz\_cov) = -4.4 \times 10^{-13}$ and numerical rank 53/74 (§5.5). On this matrix the CPU NumPy Cholesky raises `NotPositiveDefinite`, but the JAX Cholesky silently returns NaN. NUTS then receives NaN momentum and proceeds without erroring, producing the appearance of a stuck chain rather than a diagnosable failure. At minimum, `jnp.linalg.cholesky` should emit a NaN-trace warning under `jax_debug_nans` the way other NaN-producing JAX primitives do. The downstream GIGA-LENS pipeline would benefit from a defensive `cov + eps*I` jitter in any code path that feeds an SVI covariance into a TFP preconditioner.

6. **`DualAveragingStepSizeAdaptation` pathology on multimodal-Hessian models.** On this 74-parameter model the dual-averaging adapter consistently drove the step size to ∼2–3×10^{-6} within 200 burn-in steps, at which acceptance is 1.000 but per-step $\|\Delta z\|$ is 10^{-4}–10^{-3} and the chain effectively does not move (§5.5). A static log-spaced scan from 10^{-5} to 3×10^{-2} with adaptation disabled gives 0% acceptance everywhere, so the working step size lives in the narrow window $(2 \times 10^{-6}, 10^{-5})$ — below the resolution of a coarse scan and below where dual-averaging settles. We interpret this as the adapter optimizing a single global step-size for

a target acceptance under a globally Gaussian-like assumption that the model surface violates: the local Hessian at the v10 SVI mode and the local Hessian at the v11f basin differ enough that no single mass matrix or step size is simultaneously appropriate. A windowed mass-matrix adapter (e.g. `windowed_adaptive_nuts`) or a per-basin re-tuning would in principle help; both require a `JointDistribution` refactor of GIGA-Lens which we defer.

7.5 What full reproduction would require

To reproduce the published HMC posterior bit-for-bit one would need (i) the paper's exact GIGA-Lens git revision (the multi-node branch of `giga-lens/gigalens` has moved since the paper was written); (ii) mass–light position-angle tying, which is not exposed in the GIGA-Lens 2.0 public API as of our run date and would require a modification to the simulator internals; (iii) optionally a higher-order multipole correction term in the EPL profile; (iv) a working HMC at 74 parameters in JAX 0.6.2, which we did not manage to obtain (NUTS with static `max_tree_depth` resolves the JIT-compile blocker, §5.5, but step-size adaptation on this multimodal-Hessian surface remains an open problem); and (v) a `JointDistribution` refactor of the GIGA-Lens prob model to enable `windowed_adaptive_nuts` or similar window-adaptive mass-matrix samplers. These are the natural next-steps for any reader who wishes to push this reproduction further.

8 Software and Data Availability

The *HST* data used in this paper is public via MAST DOI 10.17909/hx0v-9260. The GIGA-Lens source is at `https://github.com/giga-lens/gigalens` (the `multi-node` branch at commit-time was the working baseline for this paper). The reproduction scripts (`01_download_hst.py` through `19_svi_v10_paper_mode_empirical.py` for the v1→v10 pipeline, and `20_fit_nuts_v11.py`, `21_fit_nuts_v11b.py`, `22_fit_nuts_v11c.py`, `23_nuts_stepsize_scan.py`, `24_fit_nuts_v11e.py`, `25_fit_nuts_v11f.py` for the v11 NUTS series), together with the analysis notebooks and the per-version posterior artifacts (`data/svi_v10_posterior_mass.npz`, `data/nuts_v11{,b,c,e,f}_samples.npz`, and the matching `nuts_v11*_posterior_mass.npz` files), are available on request and are planned for public release as a `foundry-i-reproduction` repository.

Table 3 lists the exact pinned versions of the software stack on the run host.

9 Conclusions

We have presented an end-to-end public-data reproduction of the GIGA-Lens modeling demonstration of Foundry I (Huang et al., 2025) on the system DESI–165.4754−06.0423. The principal findings are:

1. All six mass-model sign quadrants of the published HMC posterior are reproduced by our SVI Gaussian variational posterior.

2. θ_{E} matches the published median to 3.0%; e_1 to 2.5%; the external-shear position angle to within 1°.

3. The remaining gap — a 12° residual rotation of the lens ellipticity, $\sim 30\%$ smaller $|e_2|$ and $|\gamma_{\mathrm{ext}}|$, and a 57% steeper γ_{EPL} — is coherent rather than scattered, and is consistent with one missing physical constraint (most likely mass–light PA tying or a higher-multipole correction) rather than three independent failures.

Table 3: Software stack versions.

Package	Version
Python	3.13
JAX	0.6.2 (with jax[cuda12])
TensorFlow	2.20.0
TensorFlow Probability	0.25.0
GIGA-Lens	1.2.1 (editable; multi-node branch)
lenstronomy	1.14.0
photutils	3.0.0
astroquery	0.4.11
optax	0.2.8
NumPy	2.4.6
Astropy	current LTS
CUDA	12 (host runtime 13.0 in PATH)
Operating system	AlmaLinux 9.7 (aarch64)
GPUs used	2×NVIDIA L4 (UUID-pinned)

Note. HMC at the 74-parameter model size was not obtained under this stack; we use the v10 Gaussian SVI as the headline variational-posterior surrogate (Table 1). NUTS at the same model size *does* compile under this stack with a static `max_tree_depth` (§5.5); v11f provides a second-mode point estimate (Table 2). See §7.4 for the full upstream-issues list.

4. The Foundry I posterior surface is genuinely multimodal under the published priors: in 200 random-start MAP fits with TinyTim PSF, only 8.5% of chains land in the paper's quadrant, and the unconstrained global best lies in a different ($e_2 > 0$) mode.

5. Replacing the gigalens-default TinyTim PSF with an empirical PSF built from in-field stars flips the global-best chain into the paper's quadrant ($45/200 = 22.5\%$, with global best in mode) at the cost of a higher but more honest χ^2 floor.

6. We resolve the upstream HMC JIT-compile blocker by switching to `PreconditionedNoUTurnSampler` with a static `max_tree_depth` and no `pmap` wrapper, but the `DualAveragingStepSizeAdaptation` adapter collapses on this model: the local-curvature mismatch between modes prevents a single mass matrix or step size from working, and a windowed mass-matrix adapter would require a `JointDistribution` refactor of the GIGA-Lens prob model. The v11f NUTS run, started from the v7 paper-mode MAP point, converges into a *third* distinct local mode at $\log p \approx -63{,}868$ with $\theta_{\rm E} = 2.604''$ (1.6% off the published value, our closest reproduction of $\theta_{\rm E}$), confirming that the posterior surface is genuinely multimodal but not yielding a mixed chain.

7. We document six upstream software issues in GIGA-Lens 2.0/ JAX 0.6.2/TFP 0.25.0 that would, if resolved, materially improve the reproducibility of any Foundry-lineage analysis — notably the silent NaN from `jnp.linalg.cholesky` on a rank-deficient SVI covariance, and the dual-averaging step-size pathology on multimodal-Hessian models.

For the broader DESI Strong Lens Foundry program, and for the GIGA-Lens lineage at large, these findings argue that an in-exposure empirical PSF and a multi-start MAP scan are reasonable defaults to add to the published recipe, that mass–light PA tying or an equivalent multipole correction is a worthwhile next-order modeling improvement, that NUTS with static `max_tree_depth` should replace HMC as the default JIT-tractable sampler at this model size, and that mode-aware adaptation (windowed mass-matrix estimation, basin-conditional re-tuning, or transparent reporting of the basin a chain occupies) is a prerequisite for trustworthy posterior inference at LSST/Roman/Euclid scale.

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Software: GIGA-Lens (Gu et al., 2022), lenstronomy (Birrer and Amara, 2018), JAX (Bradbury et al., 2018), TensorFlow Probability (Dillon et al., 2017), photutils (Bradley et al., 2024), Astropy (Astropy Collaboration et al., 2022).

A Internal Ablation: v1 → v11

Table 4 traces the per-version mass-fit accuracy across the internal versions of this reproduction. The first four versions (v1–v4) explore minimal-model variants without shapelets or with unconstrained lens-light intensities. v5 introduces positive-I_e `LogNormal` priors and a tight mass-center prior, and brings θ_{E} within 0.5% of the paper. v6 adds the shapelet basis with explicit per-coefficient priors and is the smallest model that produces a physically reasonable global χ^2. v7 is a 200-chain multi-start MAP scan (TinyTim PSF) that exposes the multimodality of the surface. v8 is an SVI posterior initialised from the v7 paper-mode chain, still with TinyTim. v9 repeats v7's multi-start with the empirical PSF, flipping the global best into paper-mode. v10 is the SVI posterior initialised from the v9 global-best paper-mode chain and is the canonical headline posterior of this paper (Table 1). v11 is a NUTS-based diagnostic series (§5.5) that resolves the HMC JIT-compile blocker, exposes a silent NaN in `jnp.linalg.cholesky` on the rank-deficient SVI covariance, demonstrates that `DualAveragingStepSizeAdaptation` collapses on this model, and — in the v11f run — converges into a third local mode with $\theta_{\mathrm{E}} = 2.604''$ (1.6% off the published value, our closest single-version match for θ_{E}).

B Prior Specification

Table 5 reproduces our prior distributions, which closely follow Huang et al. (2025, Table 2) with the modifications discussed in §4.

References

Astropy Collaboration, A. M. Price-Whelan, P. L. Lim, N. Earl, N. Starkman, L. Bradley, D. L. Shupe, A. A. Patil, L. Corrales, C. E. Brasseur, et al. The Astropy Project: Sustaining and

Table 4: Ablation summary v1→v11. Best-fit mass-parameter values for each version, paper values for reference.

Version	θ_E	γ_{EPL}	e_1	e_2	$\gamma_{ext,1}$	$\gamma_{ext,2}$	χ^2_{best}
v1	+2.697	+1.66	−0.015	+0.011	+0.001	+0.053	24.65
v2	+2.062	+1.57	+0.141	−0.108	−0.001	+0.032	7.98
v4[a]	+1.523	+1.47	−0.035	−0.076	−0.036	−0.064	3.45
v5	+2.658	+1.72	−0.102	−0.179	−0.059	−0.110	7.56
v6	+2.632	+1.73	+0.128	+0.073	+0.065	+0.005	6.59
v8 (SVI/TT)[b]	+2.619	+2.20	+0.146	−0.150	+0.044	−0.106	7.72
v9 best (MAP/eP)	+2.578	+2.16	+0.103	−0.045	+0.034	−0.067	11.93
v10 (SVI/eP)[c]	+2.566	+2.15	+0.106	−0.053	+0.040	−0.068	—
v11f (NUTS/eP)[d]	+2.604	+2.26	+0.159	−0.151	+0.054	−0.119	—
Paper (HMC)	+2.6463	+1.372	+0.1091	−0.1320	+0.0657	−0.0939	—

Notes. [a] v4 has the lowest χ^2 but is unphysical: the unconstrained `use_lstsq=True` setting allows negative lens-light intensities; the resulting Einstein radius is 43% low.

[b] v8 is the first SVI posterior; initialized from the best paper-mode chain of the v7 200-chain multi-start MAP with TinyTim PSF.

[c] v10 is the canonical posterior reported in Table 1; initialized from the v9 global-best paper-mode chain (empirical PSF, $\chi^2 = 11.93$). v10 SVI does not produce a new χ^2 value.

[d] v11f is the only NUTS variant of the v11 series (§5.5) that converged into a basin distinct from the v10 SVI mode; it sits at $\log p \approx -63{,}868$ and reproduces θ_E to 1.6% but the chain did not mix and is reported as a second-mode point estimate, not a posterior. NUTS resolves the HMC JIT-compile blocker (v11) but `DualAveragingStepSizeAdaptation` collapses step size to $\sim 2\text{–}3 \times 10^{-6}$ on this model (v11a/e/f), so the chain barely moves once in a basin.

TT = TinyTim PSF. eP = empirical PSF. "Paper-mode" is the sign quadrant ($e_1{>}0, e_2{<}0, \gamma_{ext,1}{>}0, \gamma_{ext,2}{<}0$). The flip in sign quadrant at v6→v7 reflects multimodality, not a coding regression: v6 is a single-chain MAP at the global best under TinyTim, which lies in a different quadrant than the paper. v11f confirms a third distinct basin reached by gradient-based sampling.

Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935(2):167, 2022. doi: 10.3847/1538-4357/ac7c74.

S. Birrer and A. Amara. lenstronomy: Multi-purpose gravitational lens modelling software package. *Physics of the Dark Universe*, 22:189–201, 2018. doi: 10.1016/j.dark.2018.11.002.

J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs. https://github.com/google/jax, 2018.

L. Bradley, B. Sipőcz, T. Robitaille, E. Tollerud, et al. photutils: Detection, Photometry and Morphology of Sources in Astronomical Images, 2024. URL https://github.com/astropy/photutils. Astropy-affiliated package, v2.x.

A. Cikota, I. Toro Bertolla, X. Huang, S. Baltasar, N. Ratier-Werbin, W. Sheu, C. Storfer, N. Suzuki, D. J. Schlegel, R. Cartier, S. Torres, S. Cikota, and E. Jullo. DESI-253.2534+26.8843: A New Einstein Cross Spectroscopically Confirmed with VLT/MUSE and Modeled with GIGA-Lens. *The Astrophysical Journal Letters*, 2023. doi: 10.3847/2041-8213/ace9da.

DESI Collaboration, A. Aghamousa, J. Aguilar, S. Ahlen, S. Alam, et al. The DESI Experiment Part I: Science, Targeting, and Survey Design. *arXiv preprint*, 2016. doi: 10.48550/arXiv.1611.00036.

J. V. Dillon, I. Langmore, D. Tran, E. Brevdo, S. Vasudevan, D. Moore, B. Patton, A. Alemi, M. Hoffman, and R. A. Saurous. TensorFlow Distributions. *arXiv preprint*, 2017. doi: 10.48550/arXiv.1711.10604.

A. Gu, X. Huang, W. Sheu, G. Aldering, A. S. Bolton, K. Boone, A. Dey, A. Filipp, E. Jullo, S. Perlmutter, D. Rubin, E. F. Schlafly, D. J. Schlegel, Y. Shu, and S. H. Suyu. GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling. *The Astrophysical Journal*, 935(1):49, 2022. doi: 10.3847/1538-4357/ac6de4.

X. Huang, C. Storfer, V. Ravi, A. Pilon, M. Domingo, D. J. Schlegel, S. Bailey, A. Dey, R. R. Gupta, D. Herrera, S. Juneau, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.

X. Huang, C. Storfer, A. Gu, V. Ravi, A. Pilon, W. Sheu, R. Venguswamy, S. Banka, A. Dey, M. Landriau, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.

X. Huang, S. Baltasar, N. Ratier-Werbin, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2502.03455. Submitted to ApJ. 32 pages, 9 figures, 5 tables.

S. Refsdal. On the possibility of determining Hubble's parameter and the masses of galaxies from the gravitational lens effect. *Monthly Notices of the Royal Astronomical Society*, 128:307–310, 1964. doi: 10.1093/mnras/128.4.307.

W. Sheu et al. The Carousel Lens: A Well-Modeled Strong Lens with Multiple Lensed Sources. *arXiv preprint*, 2024. Citation details to be verified before journal submission.

T. Treu. Strong Lensing by Galaxies. *Annual Review of Astronomy and Astrophysics*, 48:87–125, 2010. doi: 10.1146/annurev-astro-081309-130924.

Table 5: Prior specification on the 74 free non-linear parameters of the lens model.

Parameter group / parameter	Distribution	Parameters
Mass (EPL + external shear; 8 params)		
θ_{E}	LogNormal	$\mu = \log 2.5,\ \sigma = 0.25$
γ_{EPL}	TruncatedNormal	$\mu = 2.0,\ \sigma = 0.25,\ [1.0, 2.7]$
e_1^{m}	Normal	$\mu = 0,\ \sigma = 0.1$
e_2^{m}	Normal	$\mu = 0,\ \sigma = 0.1$
c_x^{mass}	Normal	$\mu = 0,\ \sigma = 0.02\,\mathrm{arcsec}$
c_y^{mass}	Normal	$\mu = 0,\ \sigma = 0.02\,\mathrm{arcsec}$
$\gamma_{\mathrm{ext},1}$	Normal	$\mu = 0,\ \sigma = 0.05$
$\gamma_{\mathrm{ext},2}$	Normal	$\mu = 0,\ \sigma = 0.05$
Lens light — main lens (2× Sérsic, 14 params)		
R_{eff} (compact)	LogNormal	$\mu = \log 0.4,\ \sigma = 0.3$
R_{eff} (extended)	LogNormal	$\mu = \log 2.0,\ \sigma = 0.3$
n_{S}	Uniform	$[0.5, 8.0]$
$e_{1,2}^{\mathrm{light}}$	TruncatedNormal	$\mu = 0,\ \sigma = 0.15,\ [-0.4, 0.4]$
c_x, c_y	Normal	$\mu = 0,\ \sigma = 0.02\,\mathrm{arcsec}$
I_e (compact)	LogNormal	$\mu = \log 5.0,\ \sigma = 0.5$
I_e (extended)	LogNormal	$\mu = \log 2.0,\ \sigma = 0.5$
Lens light — nearby companion (2× Sérsic, 14 params)		
R_{eff}	LogNormal	$\mu = \log 0.3,\ 0.6,\ \sigma = 0.3$
c_x, c_y	Normal	$\mu = (-2.34, -2.86),\ \sigma = 0.1\,\mathrm{arcsec}$
I_e	LogNormal	$\mu = \log 1.0,\ 0.5,\ \sigma = 0.5$
e, n	As main lens	—
Source Sérsic (7 params)		
R_{eff}	LogNormal	$\mu = \log 0.5,\ \sigma = 0.3$
n_{S}	Uniform	$[0.5, 6.0]$
$e_{1,2}^{\mathrm{src}}$	TruncatedNormal	$\mu = 0,\ \sigma = 0.15,\ [-0.5, 0.5]$
c_x, c_y	Normal	$\mu = 0,\ \sigma = 0.1\,\mathrm{arcsec}$
I_e	LogNormal	$\mu = \log 2.0,\ \sigma = 0.5$
Source shapelets ($n_{\mathrm{max}} = 6,\ 1 + 2 + 28$ params)		
β	LogNormal	$\mu = \log 0.1,\ \sigma = 0.1$
c_x, c_y	Normal	$\mu = 0,\ \sigma = 0.05\,\mathrm{arcsec}$
$\mathrm{amp}_i,\ i = 0, \ldots, 27$	Normal	$\mu = 0,\ \sigma = 5/\sqrt{i+1}$

Note. The shapelet amplitude scale follows the gigalens shapelets demonstration. The companion-light centroid prior centers on the source-extraction detection on the v2 residual map (Figure 3). Total: 74 free non-linear parameters.